Evidential Actor Surfaces for Physically Consistent Driving Reconstruction

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Abstract

Reconstructing a driving scene for physical queries requires both accurate surfaces and reliable sensor returns. Sparse observations make these objectives difficult to reconcile: completing an object improves coverage but can introduce surfaces before a measured LiDAR hit. We present Evidential Actor Surfaces, an object-centric representation that learns canonical geometry and return evidence in separate stages. A four-child completion model learns from motion-compensated target sets, local surface structure, frame coverage, and differentiable ray supervision. Producer-side observations then supervise continuous free, occupied, and unknown evidence on the fixed geometry. We compose this evidence into a categorical return distribution, whose normalized measure supports an exact analysis of how component attenuation changes the probability of an early return. Separate physical, rigid-pose, and visual states provide SE(3) equivariance and isolate appearance optimization from physical geometry. On a 66-Actor nuScenes development fold, the geometry model reduces hazardous early returns by 5.12% relative, improves Chamfer distance by 6.21 mm, and increases hit recall by 2.76 percentage points. The evidential return measure further reduces hazardous early returns by 0.62 points and increases hazardous hit recall by 2.32 points over unit categorical energy on the same geometry. A frozen 352-Actor Argoverse 2 evaluation identifies the remaining cross-sensor trade-off: higher hit recall coexists with increased hazardous early returns.

1. Introduction

Reconstructed driving scenes support sensor simulation, perception analysis, and counterfactual replay. Recent neural simulators represent complex appearance and moving objects with impressive fidelity [17, 20, 24]. Physical queries impose an additional requirement: the reconstructed surface must agree with where a sensor first encounters an object. A visually plausible vehicle can still extend into measured free space, while a sparse but accurate point set can miss much of its surface. These errors affect different parts of the same ray and call for joint surface and return evaluation.

Two challenges arise in learning such a representation. First, multi-frame observations of moving Actors must be aligned before they become completion targets. A small collection of seeds also has insufficient capacity to cover a dense target surface. Second, a primitive's location and its contribution to a sensor return are different variables. The same anchor may receive occupied evidence in one view and free-space evidence in another. Assigning every overlapping Gaussian unit opacity makes the return depend on sampling density, while a single confidence score discards the structure of these observations.

We address these challenges with *Evidential Actor Surfaces* (EAS), illustrated in Fig. 1. Each tracked Actor has a canonical physical surface, continuous return evidence, and a separately owned rigid trajectory. We construct build inputs and disjoint supervision targets by transforming LiDAR sweeps into the Actor frame. A set decoder expands every completion seed into four child surfaces. Set distance, local-plane and scale targets, frame coverage, and ray losses jointly supervise the children while observed anchors stay fixed. This provides a direct route from measured geometry to the deployed surface.

For sensor termination, we retain the observation history at the corpus producer and learn separate evidence heads for anchors and children. Their free, occupied, and unknown masses summarize multi-view support and contradiction. On the fixed surface, occupied mass weights a normalized categorical distribution over ray depth. This composition yields a useful first-return measure from the learned evidence. Its normalization also permits an exact component-wise analysis: decreasing a component's weight moves the pre-hit probability toward the remaining mixture. We derive the finite change, identifying the condition under which attenuation decreases early-return probability.

The dynamic scene interface follows the same separation of variables. A read-only SE(3) pose places the canonical surface in the world; a sibling visual state carries appearance. This makes rigid composition equivariant and prevents image optimization from moving the physical surface. The representation therefore combines learnable local geometry with explicit control over which state each observation may update.

Our experiments separate geometry learning, return composition, and coordinate ownership. On the nuScenes development fold, surface supervision improves hazardous early returns, hit recall, and Chamfer jointly. Categorical evidence improves early returns and hit recall across the all, hazardous, and clear strata on the same geometry. A frozen AV2 cohort measures cross-sensor behavior, and analytical and numerical audits connect the return measure to its attenuation response. We make three contributions:

- an Actor-canonical completion representation trained with target-set, frame, and ray supervision;
- a producer-evidential categorical return measure with an exact finite attenuation identity;
- a factorized physical–pose–visual interface, evaluated through surface accuracy, first-return behavior, and SE(3) composition.

2. Related Work

Driving reconstruction and sensor returns. 3D Gaussian Splatting provides an efficient appearance representation [11]. UniSim, LiDAR4D, and DyNFL extend neural reconstruction to driving scenes and time-varying sensor observations [17, 20, 24]. LiDAR additionally measures free-space intervals before its endpoints [6]. Neural LiDAR Fields learns a ray-weight distribution for returns [7], and CaDDN represents metric depth categorically [14]. Our focus is the relationship between an Actor's completed surface and the return distribution induced by that surface. We evaluate these two objects separately and connect them through primitive-level evidence.

Completion and evidential geometry. Object-centric occupancy completion combines canonicalized LiDAR tracklets with implicit decoding [23]. PoinTr learns point-set completion, while SnowflakeNet recursively grows child points [19, 22]. Gau-Occ uses completed LiDAR geometry to initialize a Gaussian occupancy representation [12]. These approaches motivate our target-set construction and child-surface decoder. EAS further supervises canonical geometry with frame coverage and measured ray depth. ALSO and evidential occupancy methods use sensor observations to distinguish free, occupied, and unobserved space [1, 9, 10]. GaussianFormer-2, GaussRender, and Vol3DGS relate Gaussian support to occupancy or ray transmittance [3, 8, 15]. We study how continuous evidence should compose on a fixed surface, comparing additive transmittance with categorical return mass.

Motion and appearance factorization. Motion compensation is essential when constructing targets from accumulated sweeps [18]. DualAD separates static and dynamic streams [4]; DynamicVGGT predicts current and future point maps and supervises Gaussian motion with scene flow [5]. Neural scene graphs provide object transforms for dynamic reconstruction [13]. EAS uses annotated rigid motion to canonicalize evidence and relocate the learned surface. Its visual state follows the attribute-on-geometry principle explored by Feature 3DGS [25], with a stop-gradient physical carrier that gives appearance updates a separate owner.

3. Problem Formulation

For tracked Actor i, let $Z_i = (ID_i, T_{i,0:H}, D_i, C_i)$ denote identity, rigid trajectory, dimensions, and category. Build observations B_i contain LiDAR endpoints and their sensor origins. Disjoint target sweeps Y_i supply surface and first-return labels. We learn a canonical physical state $P_i = (S_i, m_i)$, where S_i contains primitive centers and support scales, and m_i contains continuous evidence. The Actor state Z_i is preserved during reconstruction.

For a target ray $r = (o, v, d^\star)$, a return $\widehat{d}$ is early when $\widehat{d} < d^\star - \tau$ and a hit when $|\widehat{d} - d^\star| \le \tau$. We use $\tau = 0.20\,\mathrm{m}$. The literal point-surface operator takes the smallest positive projected depth of any center within a $0.20\,\mathrm{m}$ lateral beam tube; an empty tube has depth ∞. The categorical operator instead returns the interpolated median of the surface's depth distribution. Both are scored against the same measured endpoint.

The learning objective combines surface reconstruction with ray agreement. Evaluation reports early rate, hit recall, and symmetric Chamfer together: each measures a distinct consequence of changing S_i. An Actor's hazard label is used only for stratified evaluation. A read-only pose places P_i in world coordinates, and a separately parameterized visual state V_i supplies appearance.

4. Method

4.1. Motion-Compensated Surface Supervision

Annotated sensor, ego, and Actor poses map each endpoint into a canonical frame,

$$x_{it} = T_{it}^{-1} T_{\mathrm{ego},t} T_{\mathrm{sensor}} x_t^{\mathrm{sensor}}. \tag{1}$$

We transform its sensor origin by the same chain. Build sweeps produce fixed observed anchors $\mathcal{A}_i$ and completion seeds; separate sweeps produce the target set $\mathcal{Y}_i$. Observed anchors comprise retained endpoints and endpoints projected onto build-supported geometry.

A pretrained relocation model supplies parent center $\bar{x}_j$. A set decoder combines build features with four learned slot embeddings and predicts bounded residuals and isotropic scales:

$$c_{jk} = \bar{x}_j + \Delta x_{jk}, \quad S_i = \mathcal{A}_i \cup \{(c_{jk}, s_{jk}) : k = 1, \dots, 4\}. \tag{2}$$

Residuals are bounded by $0.25\,\mathrm{m}$ per coordinate and scales lie in $[0.02, 0.40]\,\mathrm{m}$. We write $\mathcal{X}_i$ for the centers of S_i. The

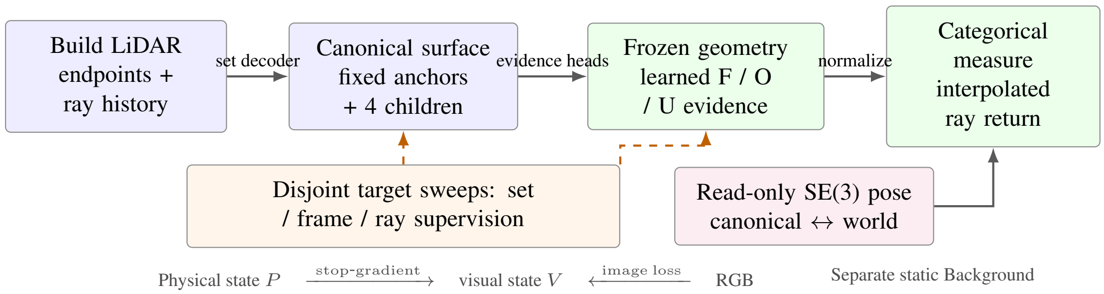

Figure 1. Overview of Evidential Actor Surfaces. Motion-compensated build observations produce a canonical surface. Disjoint targets supervise geometry and continuous free/occupied/unknown (F/O/U) evidence in stages. Frozen evidence weights a categorical ray measure. Rigid pose and appearance have separate state ownership. Dashed arrows denote supervision.

decoder uses Actor dimensions and canonical geometric evidence. Its target is a set, allowing several children to cover distinct parts of a local surface.

Geometry and frame coverage. Let $\widetilde{\mathcal{L}}$ denote a loss normalized by its detached parent-reference value. We combine symmetric nearest-neighbor distance, local point-to-plane distance, and log-scale regression:

$$\mathcal{L}_G = \widetilde{\mathcal{L}}_{\text{set}} + .25\widetilde{\mathcal{L}}_{\text{plane}} + .10\widetilde{\mathcal{L}}_{\text{scale}} + \widetilde{\mathcal{L}}_{\text{frame}}. \quad (3)$$

Local planes and scale targets come from eight target neighbors. The frame term averages directed target-to-surface distance equally over target frames,

$$\mathcal{L}_{\text{frame}} = \frac{1}{|\mathcal{F}_i|} \sum_{t \in \mathcal{F}_i} \frac{1}{|\mathcal{Y}_{it}|} \sum_{y \in \mathcal{Y}_{it}} \min_{x \in \mathcal{X}_i} \|y - x\|_2. \quad (4)$$

This prevents densely sampled frames from dominating canonical coverage.

Differentiable ray supervision. We sample 48 depths along each training ray and alpha-compose Gaussian support into an expected depth $\bar{d} = \sum_k T_k \alpha_k d_k + T_{\text{end}} d_{\text{far}}$. Here $d_{\text{far}} = d^\star + 0.50\,\text{m}$ supplies the residual no-hit depth. The ray objective is

$$\mathcal{L}_R = \widetilde{\text{Huber}}(\bar{d}, d^\star) + .5\,[\widetilde{d^\star - \tau} - \bar{d}]_+. \quad (5)$$

The second term penalizes a predicted depth inside the measured pre-hit interval. This is a smooth depth surrogate; the literal beam-tube minimum is used for evaluation. During physical fine-tuning, PCGrad [21] projects conflicting geometry and ray gradients. All children are emitted at deployment, followed by a fixed $0.06\,\text{m}$ voxel deduplication.

4.2. Producer-Evidential Return Measure

We next fix the geometry and learn how its primitives contribute to sensor termination. For each anchor, the producer retains source-frame and ray provenance, projection displacement, canonical hit count, temporal/view support, and continuous build evidence. Children inherit parent evidence together with their local geometry. Separate PointNet-style heads predict $m_j = (m_{F,j}, m_{O,j}, m_{U,j})$ by a three-way softmax. Target-ray votes supervise these masses with soft cross-entropy: observations before a hit contribute free evidence, endpoints contribute occupied evidence, and unsupported or occluded regions contribute unknown evidence.

For a completion child, let n_j count positive-depth ray opportunities inside the beam tube, and let $n_{F,j}$ and $n_{O,j}$ count pre-hit and endpoint votes. Its target is

$$m_j^\star = \left(\frac{n_{F,j}}{n_j}, \frac{n_{O,j}}{n_j}, 1 - \frac{n_{F,j} + n_{O,j}}{n_j} \right), \quad n_j > 0, \quad (6)$$

with $(0, 0, 1)$ when no ray supports the child. Both free and occupied coordinates can be positive, preserving disagreement across observations. The evidence loss is $\mathcal{L}_E = -\sum_j \sum_{a \in \{F,O,U\}} m_{a,j}^\star \log m_{a,j}$, averaged over the supervised primitives.

Staged evidence learning. The anchor head is trained with evidential and ordered-transmittance return losses. With that head frozen, the child head is trained with the same objectives and then fine-tuned with a pre-hit survival term, the integrated child optical thickness before $d^\star - \tau$. These stages learn evidence on fixed centers and scales. We subsequently freeze both heads and change the return composition to a categorical measure. Thus the final composition is a frozen representation experiment; it does not require joint categorical fine-tuning.

Categorical composition. For 64 ordered samples d_k inside the Actor box, define

$$\kappa_{kj} = \exp\left(-\frac{\|o + d_k v - c_j\|_2^2}{2 s_j^2} \right), \qquad w_j = m_{O,j}. \quad (7)$$

The joint depth–primitive distribution is

$$q_{kj} = \frac{w_j \kappa_{kj}}{\sum_{a,b} w_b \kappa_{ab}}, \qquad p_k = \sum_j q_{kj}, \quad F_k = \sum_{\ell \le k} p_\ell. \tag{8}$$

We interpolate within the first CDF bin crossing $1/2$ to obtain the return $\widehat{d}$. The measure conditions on a return inside the annotated box. Multiplying all primitive weights by the same positive constant leaves p_k unchanged. This removes a global optical-thickness degree of freedom; changing the relative mass of a primitive family still changes the distribution.

4.3. Component Responsibility and Attenuation

For an evaluation boundary $b = d^\star - \tau$, interpolate $F(b)$ with the same bins as the median. At interior boundaries with positive bin mass,

$$\widehat{d} < b \quad \iff \quad F(b) > \tfrac{1}{2}. \tag{9}$$

Because the measure is additive over primitives, $F(b) = F_{\mathcal{A}}(b) + F_{\text{child}}(b)$. This decomposes the deployed early event into observed-anchor and completion contributions.

Let $D_j = \sum_k \kappa_{kj}$, let $N_j(b)$ be the interpolated pre-boundary kernel mass, and define $C_j = N_j/D_j$ and $r_j = w_j D_j / \sum_a w_a D_a$. Then

$$\frac{\partial F(b)}{\partial \log w_j} = r_j(C_j - F(b)). \tag{10}$$

For a finite attenuation $w_j \mapsto v w_j$, $0 < v < 1$,

$$F_v(b) - F(b) = \frac{(1-v) r_j \big(F(b) - C_j\big)}{1 - (1-v) r_j}. \tag{11}$$

The denominator is positive. For a component of positive mass, attenuation decreases $F(b)$ exactly when $C_j > F(b)$; it increases $F(b)$ when $C_j < F(b)$. The same identity holds for uniform attenuation of a family after aggregating its mass. Its sign is independent of attenuation magnitude for that fixed component or family. Indeed, the attenuated measure is $F_v = (F - (1-v) r_j C_j)/(1-(1-v) r_j)$; subtracting F gives Eq. 11. We use target-defined b for analysis.

4.4. Rigid Composition and Visual Ownership

The canonical energy $e_i(x) = \log \sum_j w_j \exp(-\|x - c_j\|^2/(2 s_j^2))$ is placed in the world by

$$Q_{it}(x) = e_i\big(R_{it}^\top (x - t_{it})\big), \qquad T_{it} = (R_{it}, t_{it}). \tag{12}$$

For a common rigid transform g, inverse querying gives $Q_{gT}(gx) = Q_T(x)$. Rigid pose changes location while preserving canonical pairwise distances. Static Background has a separate scene owner.

The visual layer receives a detached physical carrier $\mathrm{sg}(P_i)$ and owns its render geometry, spherical harmonics, and opacity. Image losses update only this sibling state. Since the physical query reads (P_i, T_{it}), its derivative with respect to visual parameters is zero. This interface lets appearance capacity grow independently of the physical surface.

5. Experiments

5.1. Experimental Setup

Data. We use nuScenes [2] for learning and Argoverse 2 Sensor validation [16] for the frozen external evaluation. The geometry and evidence stages use 593 training Actors and a 66-Actor development fold (41 hazardous, 25 clear; 99,208 rays). The fold was exposed during predecessor development, so source results characterize the learned mechanism. AV2 selection excludes the 60 logs used by the earlier compiler studies from 150 validation logs. Positions $0, 4, \ldots, 76$ in the sorted 90-log complement give 20 logs, 352 Actors, and 1,016,652 rays. Models, transforms, and evaluation parameters are frozen before this evaluation.

Metrics and implementation. Early and hit rates use all target rays as their denominator; misses remain in that denominator. Chamfer is the Actor-mean symmetric nearest-neighbor distance. We report all, hazardous, and clear strata. Surface comparisons use literal beam-tube returns; evidence comparisons use interpolated categorical returns on identical geometry. All Actor states are retained. The four-child decoder has hidden width 128 and 16-dimensional slots; frame-aware fine-tuning runs six epochs with AdamW, learning rate 10^{-4}, and four Actors per batch. Evidence heads use hidden width 64. The local archive gives their staged training schedule and run identities.

5.2. Canonical Surface Reconstruction

Table 1 compares the original completion surface with target-set expansion and its frame-balanced extension. Set expansion reduces hazardous early returns from 27.80% to 24.96%, improves Chamfer by 2.90 mm, and increases hit recall by 2.17 points. Frame coverage further improves Chamfer to 231.46 mm and hit recall to 50.43%. It also reduces frame-mean target distance by 7.01 mm for moving Actors and 3.55 mm for quasi-static Actors relative to set expansion. Its hazardous early rate is 26.37%, a 5.12% relative reduction from the original surface.

Frame coverage therefore improves the completeness side of the trade-off while retaining the hazardous early-return improvement over the baseline. Clear early returns rise from 21.56% to 22.39%; the stratified table makes this trade-off explicit. Figure 2 visualizes set expansion on an

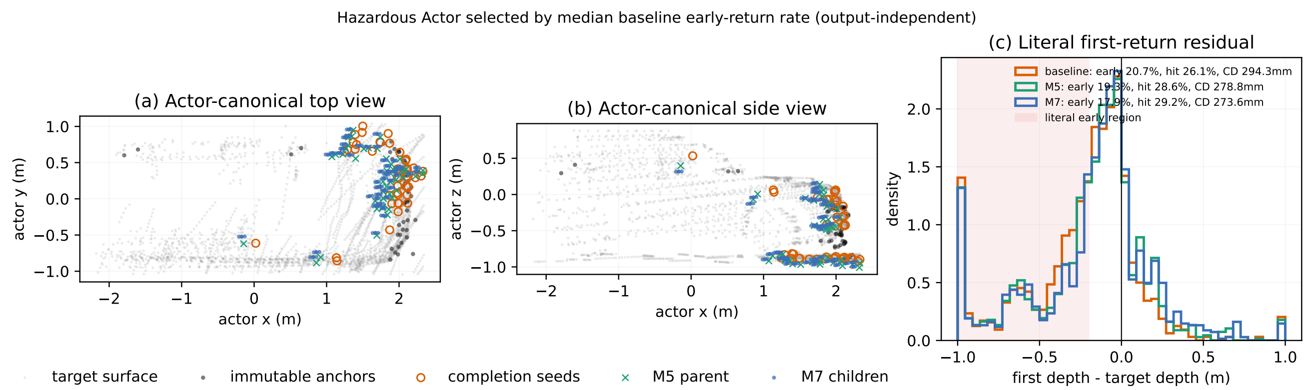

Figure 2. Canonical set expansion on a hazardous development Actor. The 47 completion seeds produce 188 children while observed anchors stay fixed. Original/relocation/set-expansion early rates are 20.7/19.3/17.9%, hit rates are 26.1/28.6/29.2%, and Chamfer distances are 294.3/278.8/273.6 mm. Gray points are disjoint target observations. This example shows the set-expansion stage before frame fine-tuning.

Table 1. Canonical geometry on 66 development Actors. Early and hit are ray percentages; CD is Actor-mean Chamfer in mm. All rows use literal point-surface returns and retain 100% of Actor states.

Method	Early ↓ All	Hazard	Clear	CD ↓ (mm)	Hit ↑ (%)
Original	26.74	27.80	**21.56**	237.67	47.67
Set expansion	**24.41**	**24.96**	21.74	234.77	49.84
+ frame coverage	25.70	26.37	22.39	**231.46**	**50.43**

Actor chosen by its baseline early-return median before inspecting learned output. The additional children cover the held-out surface more closely.

5.3. Evidence and Return Composition

Producer replay aligns 297,535 anchors across 1,004 corpus Actors with build and target ray evidence. Build and target occupied masses correlate at $r = 0.514$, while 58.72% of anchors receive both free and occupied target votes. This supplies a predictive but contradictory observation history for the evidence heads.

Table 2 isolates the effect of learned evidence on fixed frame-supervised geometry. The categorical measure lowers early rate from 18.19% to 17.67% and raises hit recall from 62.28% to 64.46%. Hazardous early rate decreases by 0.62 points and hazardous hit recall increases by 2.32 points. All three strata improve both metrics. The clear early decrease is small (0.03 points); the larger gains occur in hit recall.

Why the composition matters. Using the same learned anchor and child evidence as additive transmittance gives 21.51% early returns and 66.92% hit recall. Categorical

Table 2. Categorical return comparison on identical geometry within each dataset. Unit: unit Gaussian energy. EAS: learned evidential weights. All values are percentages; source and AV2 are distinct cohorts.

Stratum	Early ↓ Unit	EAS	Hit ↑ Unit	EAS
nuScenes development: 66 Actors / 99,208 rays				
All	18.19	**17.67**	62.28	**64.46**
Hazardous	17.42	**16.80**	62.07	**64.39**
Clear	21.95	**21.92**	63.34	**64.80**
AV2 frozen: 352 Actors / 1,016,652 rays				
All	16.91	17.14	38.35	44.24
Hazardous	16.56	17.10	27.98	34.28
Clear	17.21	17.17	47.14	52.68

composition shifts this operating point to 17.67%/64.46%. The ordered optical model is therefore useful for learning evidence, but its accumulated front-tail mass produces earlier returns on this primitive set. An exact responsibility audit finds that the categorical model reduces mean pre-boundary mass by 0.331 points: anchors contribute -0.346 points and children $+0.016$ points. The observed-anchor evidence accounts for most of the net reduction.

5.4. Attenuation and First-Return Behavior

Figure 3 evaluates Eq. 11 on 99,208 frozen rays. Uniform child attenuation has the adverse sign on 58.85% of rays and the favorable sign on 30.96%. For a learned component-wise visibility ablation, exact pre-boundary mass increases on 59.32% of rays and decreases on 27.27%. The first-order prediction agrees with the finite-change sign on 95.73%. These observations illustrate the distinction between reducing a component's weight and reducing the normalized mixture's probability before a boundary.

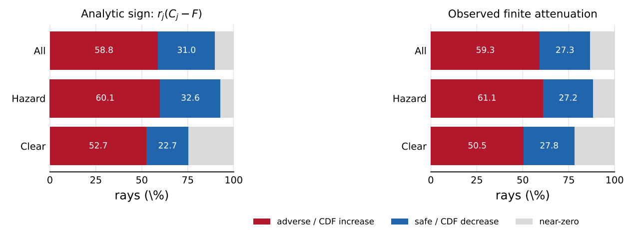

Figure 3. Attenuation response on frozen development rays. Left: sign predicted for uniform child attenuation. Right: exact change under the learned visibility ablation. Red indicates increased pre-boundary probability; blue indicates a decrease.

The smooth ray loss has a related operator boundary: expected-depth error does not determine the literal minimum over supported centers. In a frame-balanced ray-loss ablation, 38.03% of newly early rays improve their smooth depth error even when both operators use exactly the same support. The local archive gives a constructive counterexample and the full same-support/deployment comparison.

5.5. Dynamic Scene Composition

We match 12 scene-0230 Actors to a read-only StreetGS checkpoint. Across 36 Actor–frames, 5,791 physical Gaussians follow rigid poses independently of 106,807 appearance-owned and 1,140,862 Background Gaussians. The unit-weight composition audit yields maximum energy and pairwise-distance residuals of 3.40×10^{-14} and 1.24×10^{-14} m under translations up to 126.28 m. This verifies the coordinate composition in Eq. 12. Separate visual-parameter optimization leaves physical centers and scales fixed, validating the gradient ownership used by the scene interface. The archive reports the associated appearance-capacity experiments.

5.6. Cross-Sensor Evaluation

On the frozen AV2 cohort, the categorical model increases overall hit recall from 38.35% to 44.24% and hazardous hit recall from 27.98% to 34.28% (Table 2). Early rates rise by 0.23 points overall and 0.54 points for hazardous Actors. The all-early and worst-stratum nonincrease criteria consequently fail, while hit retention passes. The complete result shows a source-to-target trade-off: the return measure recovers more measured endpoints, but its source-domain early-return improvement does not transfer. No AV2 fine-tuning, calibration, or threshold adjustment is applied.

Scope and limitations. The present model addresses rigid, annotated Actors with sufficient multi-frame LiDAR support and conditions on a return inside the Actor box. Sparse sensing and pose errors remain sources of surface ambiguity. Source results use an exposed development fold; the frozen AV2 evaluation quantifies the remaining sensor dependence. Non-rigid deformation, explicit ray drop, and static-world completion require additional targets and modeling.

6. Conclusion

We presented Evidential Actor Surfaces, a staged representation for canonical geometry and sensor termination. Target-set and frame supervision improve surface reconstruction, while producer evidence supplies a useful categorical return measure on fixed geometry. The finite attenuation identity characterizes how component weights affect the deployed early-return boundary, and typed state ownership preserves rigid equivariance during appearance optimization. Together, the experiments connect physical reconstruction quality to its supervision, return operator, and coordinate interface.

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